Set up LAN

We started by getting the light blue cable that connects the computer to the switch, as well as the power cable that turns on the switch. We plugged the power switch cord to turn on the switch, then I plugged the light blue cable in so that my computer is able to communicate with the network switch. We have to plug the computer into the switch because the switch by itself doesn't have a screen, so we used the computer and a program called Tera Term to act as the switch's screen. At the end of this process, we had to make sure that the serial setup was correct by checking whether there was a 9600 baud rate so that the computer and switch could communicate properly. The next step we had to take was to erase all the old configurations and VLANs from the database so that when we were ready to enter ours, the old data wouldn't overlap with the new data we put in. We loaded the switch back up so it could restart without the old data. For step 3, we assigned an IP address to the switch so we could manage and communicate with it to specify which functions to run. In step 4, we created 5 VLANs, or 5 departments, within a business; even though these VLANs are connected to one switch, they are organized in different sections or networks so that one department does not have another department's information and keeps everything secure. In step 5, we prevented unauthorized users from gaining access by adding port security to our ports that only allow a max of 1 MAC address. The switch learned and remembered the MAC address because we added the sticky command, and if an unauthorized device tries to connect to the switch, the restrict command kills its attempt.

In step 6, we created an MOTD banner for unauthorized users trying to connect to the switch, saying that legal action will be taken if you try to access this switch, and then we saved the configuration so that everything is saved even after a restart. In step 7, we created a system that allows us to control the switch remotely by setting up SSH and changing the hostname to SW1 and the domain name to atlanta-campus. Locally, create an encryption key that helps secure the device's communication and then set up an account. The last thing we did was create VTY lines to connect with SSH so it would be more secure. In step 8, we tried IP routing so that the different devices in the VLANs can communicate on the switch. Then we created and assigned IP addresses for every VLAN 10-50 (e.g., 192.168.10.1) so that the traffic can be routed between the different VLAN networks. In step 9, we wanted to stop the sales VLAN from communicating with the executive VLAN, so we created secure_execs, which is an access list that restricts the ability of the sales VLANs to talk to the executive one.

In step 10, we wanted the switch to automatically give the computers that are connected their IP address and the DNS server information, so we enabled the DHCP server for each VLAN. In step 11, we divided 20 physical switch ports evenly to make sure that the DHCP pools are working; we divided them up among the 5 departments. For example, we assigned ports 1-4 to IT and 5-20 to the other departments. To check, we moved ports and renewed the IP address to make sure everything was running correctly. In step 12, we used TFTP to transfer all the settings that were being used and put them on another computer. This means we don't have to reconfigure everything, and if the switch loses its settings, we can restore them. In step 13, we made the switch time accurate by adding an NTP server and a logging host, which allows the

switch without physically being near or close to the switch or being remote. We also added PortFast and BPDU Guard for faster access and more security against unwanted devices. The CPD was disabled so that there is little to no information exposure. Unused ports were shut down and put into Blackhole so the unwanted devices cannot be plugged in and used.